CASO GENERAL CONVECC-DIFUSION

AIRE REPOSO = SOLIDO $\vec{u}(u_1,u_2)=0$

en que particulas de Ω se mueven (*)

(1) $$\boxed{a\frac{Du}{Dt}} - K\left[\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right] = f$$

$$\parallel$$

$$\boxed{a\left[\frac{\partial u}{\partial t} + u_1\frac{\partial u}{\partial x} + u_2\frac{\partial u}{\partial y}\right]} \quad (2)$$

si $u \equiv$ temperatura

$\Downarrow$

$a = \rho \cdot c$

K = conduct. termica

(*) $\vec{u}(u_1,u_2)$ velocidad

$\longrightarrow$ si $u_1=u_2=0$ $\vec{u}=0$ CASO SOLIDO, O FLUIDO en REPOSO

MEF:

$$\underbrace{a\frac{\partial u}{\partial t} - \underbrace{K\left[\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right]}_{\text{caso estacionario}} = f}_{\text{caso evolutivo}}$$

(1) $$\boxed{\int_\Omega a\frac{Du}{Dt}\, v\, d\Omega} + \int_\Omega \left(K_x\frac{\partial u}{\partial x}\frac{\partial v}{\partial x} + K_y\frac{\partial u}{\partial y}\frac{\partial v}{\partial y}\right) d\Omega - \sum_i \int_{\Gamma_i} \underbrace{\left(K\frac{\partial u}{\partial n}\right)}_{\sigma_n} v\, d\Gamma$$

dos posibilidades:

1a) directamente, en free fem $\boxed{\frac{Du}{Dt}} \simeq \boxed{\frac{u^{n+1}(P,t_{n+1}) - \text{convect}(P,t_n)}{\Delta t}}$ posición donde mano partícula en t_n

1b) modo estandard sustituyendo (2)

$$\int_\Omega a\frac{Du}{Dt}\, v\, d\Omega \longrightarrow \int_\Omega a\underbrace{\frac{\partial u}{\partial t}}_{=\frac{u^{n+1}-u^n}{\Delta t}} v\, d\Omega + a\int_\Omega \left(u_1\frac{\partial u}{\partial x}v + u_2\frac{\partial u}{\partial y}v\right) d\Omega$$

$u \equiv$ puede ser cualq. variable escalar, ej: concentrac. contaminante, un potencial,